Anticoagulación en fibrilación auricular

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IMPROBABLE RESEARCH

Research that makes people LAUGH...then THINK

RESEARCH ARTICLE

Cows painted with zebra-like striping can avoid biting fly attack

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Med

CellPress

Clinical and Translational Resource and Technology Insights

Mammalian enteral ventilation ameliorates respiratory failure

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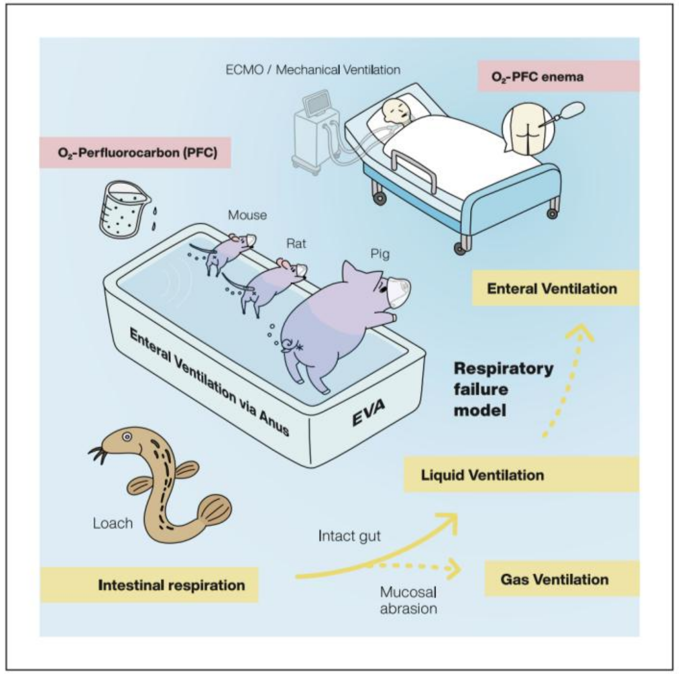

Ryo Okabe, Toyofumi F. Chen-Yoshikawa, Yosuke Yoneyama, ..., Eiji Kobayashi, Hiroshi Date, Takanori Takebe

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Highlights

Enteral ventilation (EVA) enabled systemic oxygenation in mammalians

EVA improved survival and behaviors in pre-clinical models of respiratory failure

EVA yielded no major signs of complications in pre-clinical models

EVA may serve as a transformative approach in respiratory failure patients

DOI: 10.1111/ecot.12259

ORIGINAL ARTICLE

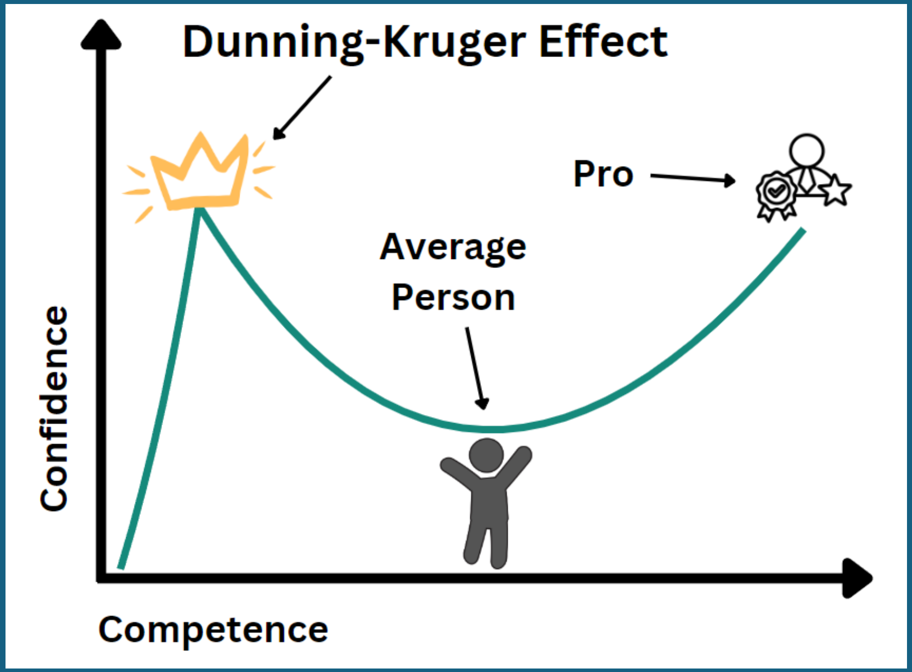
ECONOMICS OF TRANSITION AND INSTITUTIONAL CHANGE

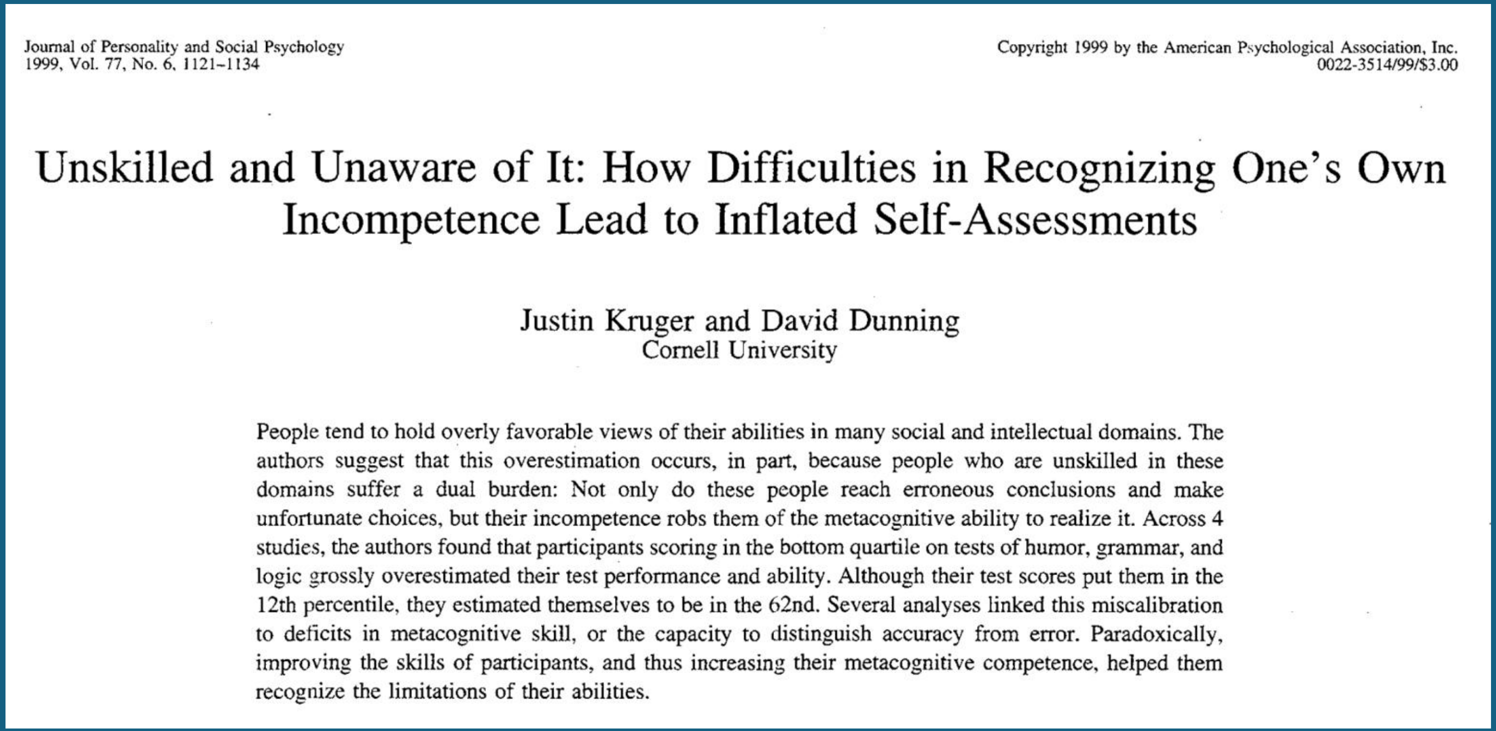
WILEY

Obesity of politicians and corruption in post-Soviet countries

Pavlo Blavatskyy

Country	Median estimated ministers' BMI	Image of a minister with median BMI	Corruption Perceptions Index 2017	World Bank Control of Corruption 2017	Index of Public Integrity 2017	IDEA Absence of Corruption 2017	Basel AML Index 2017
Estonia	28.7		71	1.24	8.93	0.807	2.73
Lithuania	30.3		59	0.55	7.82	0.641	3.67
Latvia	30.7	 	58	0.54	7.93	0.631	3.64
Georgia	30.9		56	0.74	7.30	0.652	5.28
Armenia	32.1		35	−0.56		0.352	4.44
Russia	32.5		29	−0.89	5.90	0.249	5.7
Moldova	32.7		31	−0.80	6.44	0.370	5.23





Journal of Personality and Social Psychology
1999, Vol. 77, No. 6, 1121–1134

0022-3514/99/$3.00

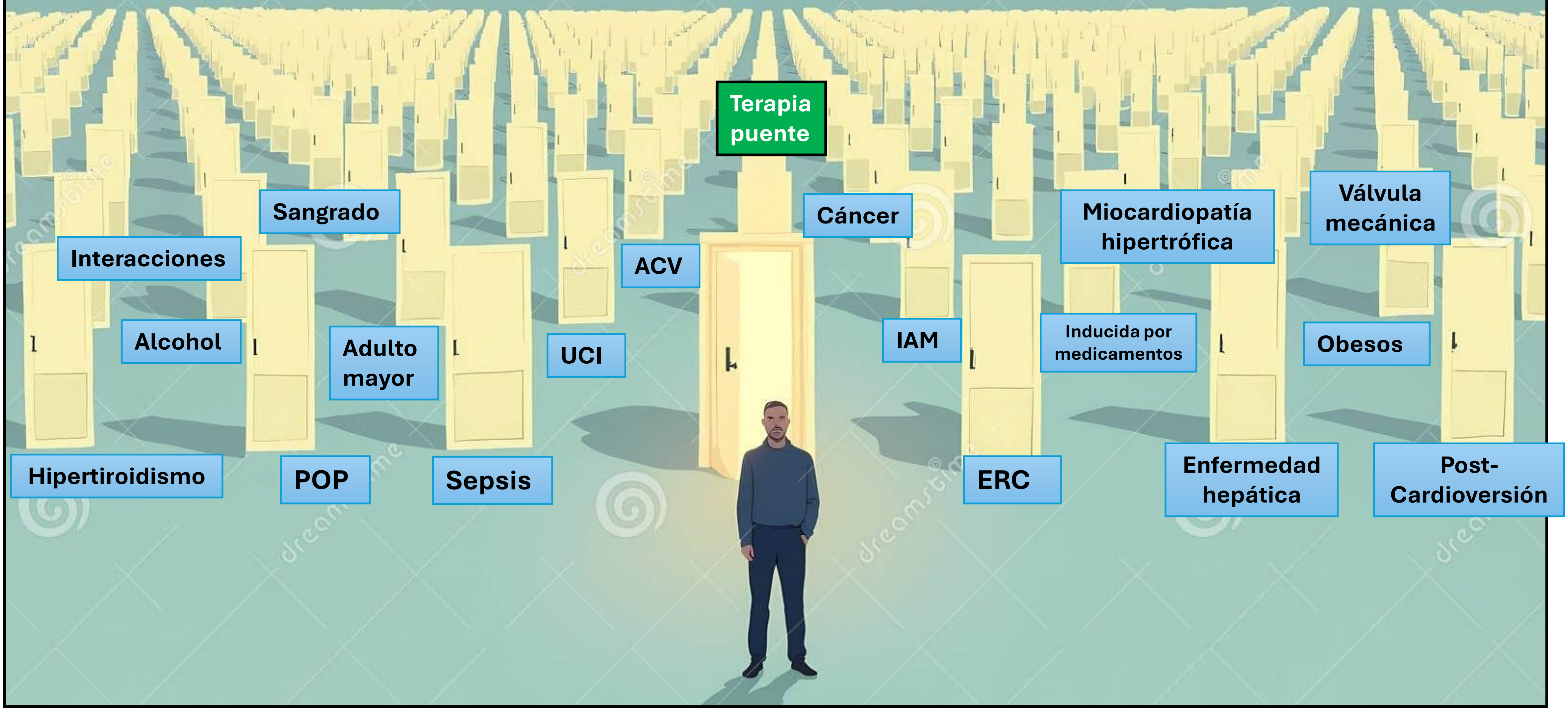
Unskilled and Unaware of It: How Difficulties in Recognizing One's Own Incompetence Lead to Inflated Self-Assessments

Justin Kruger and David Dunning
Cornell University

People tend to hold overly favorable views of their abilities in many social and intellectual domains. The authors suggest that this overestimation occurs, in part, because people who are unskilled in these domains suffer a dual burden: Not only do these people reach erroneous conclusions and make unfortunate choices, but their incompetence robs them of the metacognitive ability to realize it. Across 4 studies, the authors found that participants scoring in the bottom quartile on tests of humor, grammar, and logic grossly overestimated their test performance and ability. Although their test scores put them in the 12th percentile, they estimated themselves to be in the 62nd. Several analyses linked this miscalibration to deficits in metacognitive skill, or the capacity to distinguish accuracy from error. Paradoxically, improving the skills of participants, and thus increasing their metacognitive competence, helped them recognize the limitations of their abilities.


DOAC vs WARFARINA
Misma meta: prevenir el ACV en FA
Rápido, predecible y sin tantas restricciones
Inicio rápido de acción
No requiere monitoreo rutinario
Menos interacciones con alimentos
Menos interacciones medicamentosas
Menor riesgo de hemorragia intracraneal
Adherencia clave: una dosis olvidada reduce la protección
DOAC
Apixabán
Rivaroxabán
Dabigatrán
Edoxabán
Veterana, económica y aún esencial en situaciones clave
INR
Requiere monitoreo (INR 2-3)
Interacciones con alimentos (vitamina K)
Muchas interacciones medicamentosas
Inicio y offset más lentos
Mayor riesgo de hemorragia intracraneal
Económica y de amplio acceso
Antídotos ampliamente disponibles
WARFARINA
Antagonista de la vitamina K
RECORDAR
Evaluar siempre:
• Riesgo de ACV (CHA_2DS_2-VASc)
• Riesgo de sangrado (HAS-BLED)
• Función renal
• Preferencias del paciente
IDEAL PARA PACIENTES QUE:
✓ Prefieren comodidad y no controles frecuentes
✓ Tienen riesgo de HIC
✓ No tienen válvula mecánica ni estenosis mitral moderada-severa
✓ Pueden costear / acceder al tratamiento
INDIVIDUALIZAR ES LA CLAVE
IDEAL PARA PACIENTES QUE:
✓ Tienen válvula mecánica o estenosis mitral moderada-severa
✓ Tienen enfermedad renal avanzada (TFG <15 ml/min o diálisis)*
✓ Necesitan un anticoagulante con antídoto accesible
✓ Tienen limitaciones económicas

Escenarios de cada día...

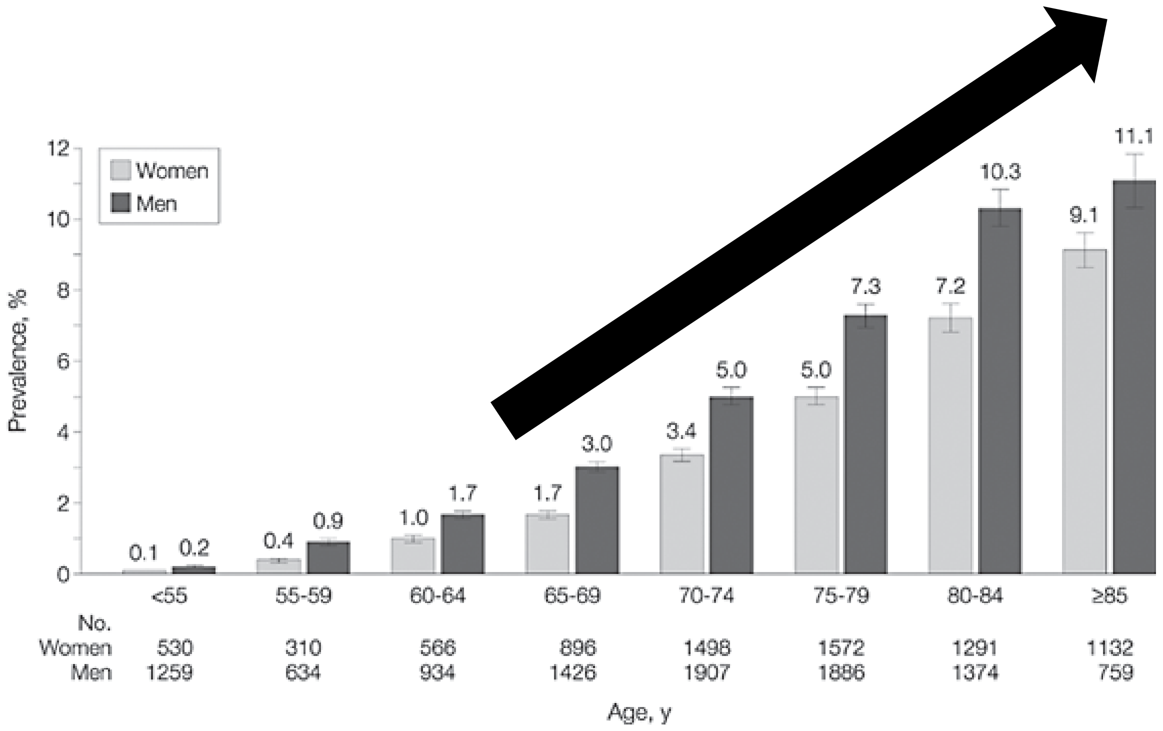


Contenido

1. Importancia de la anticoagulación en fibrilación auricular.
2. Características principales de los anticoagulantes orales.
3. Evaluación del riesgo tromboembólico y de sangrado.
4. Indicación de la terapia anticoagulante.
5. Escenarios especiales.

Importancia de la anticoagulación

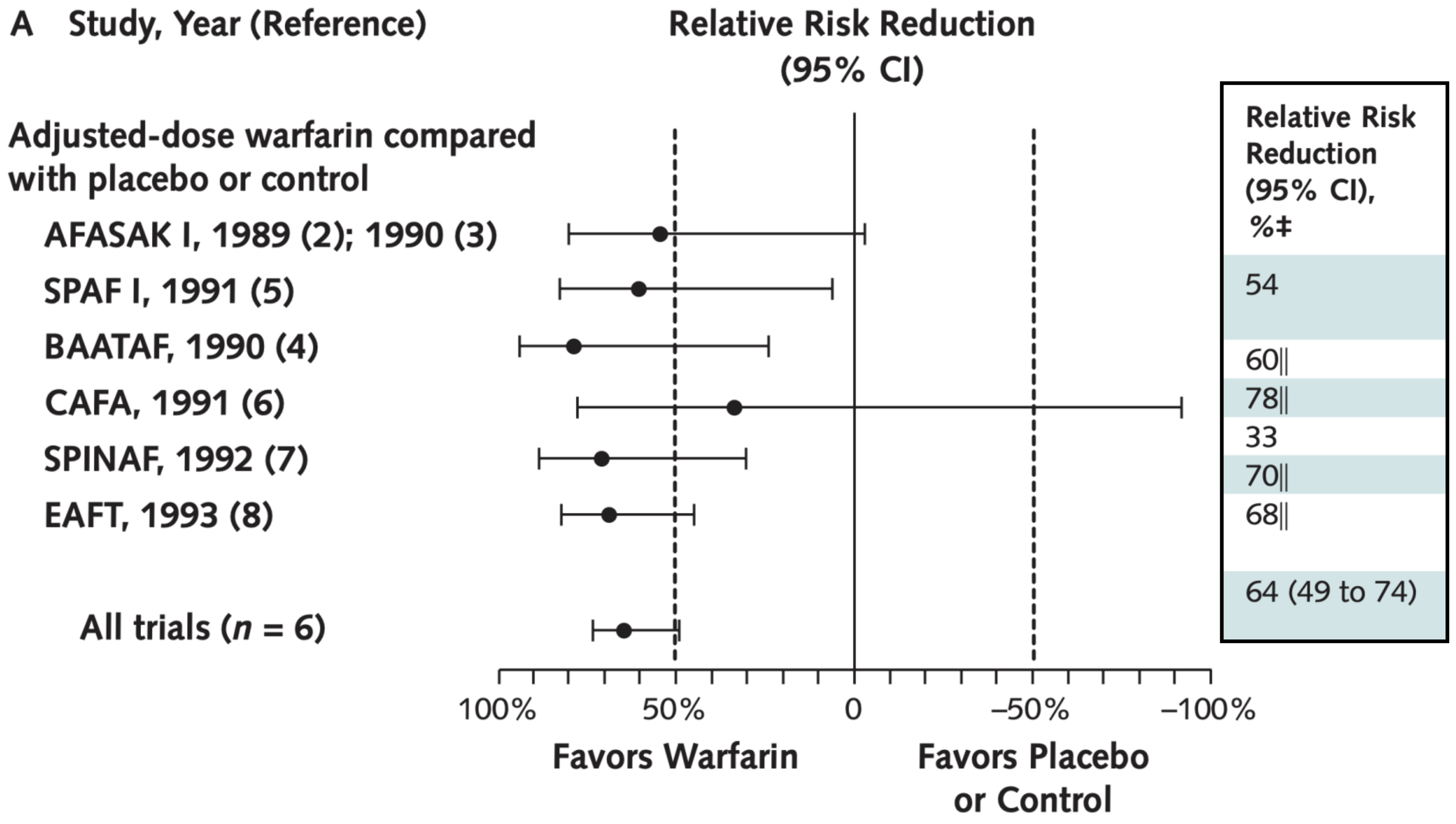
Comportamiento de la enfermedad



- ✓ *1 de cada 4 adultos mayor de 40 años va a desarrollar FA.*
- ✓ *De esos, el 35% presentará un ACV.*
- ✓ *Doble de riesgo de mortalidad.*
- ✓ *5 veces más riesgo de HF y stroke.*
- ✓ *Efecto mortal o incapacitante con primer evento.*

Go AS, Hylek EM, Phillips KA, et al. Prevalence of Diagnosed Atrial Fibrillation in Adults: National Implications for Rhythm Management and Stroke Prevention: the AnTicoagulation and Risk Factors In Atrial Fibrillation (ATRIA) Study. JAMA. 2001;285(18):2370–2375.

Impacto de la anticoagulación con Warfarina

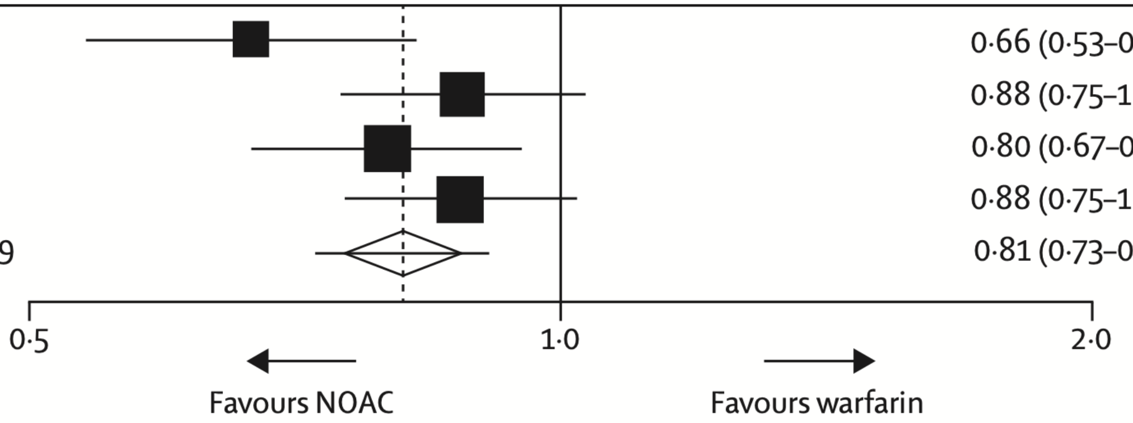


Warfarina vs. Placebo: Reducción del **64%** del riesgo de ACV.

Hart RG, Pearce LA, Aguilar MI. Meta-analysis: antithrombotic therapy to prevent stroke in patients who have nonvalvular atrial fibrillation. Ann Intern Med. 2007 Jun 19;146(12):857-67.

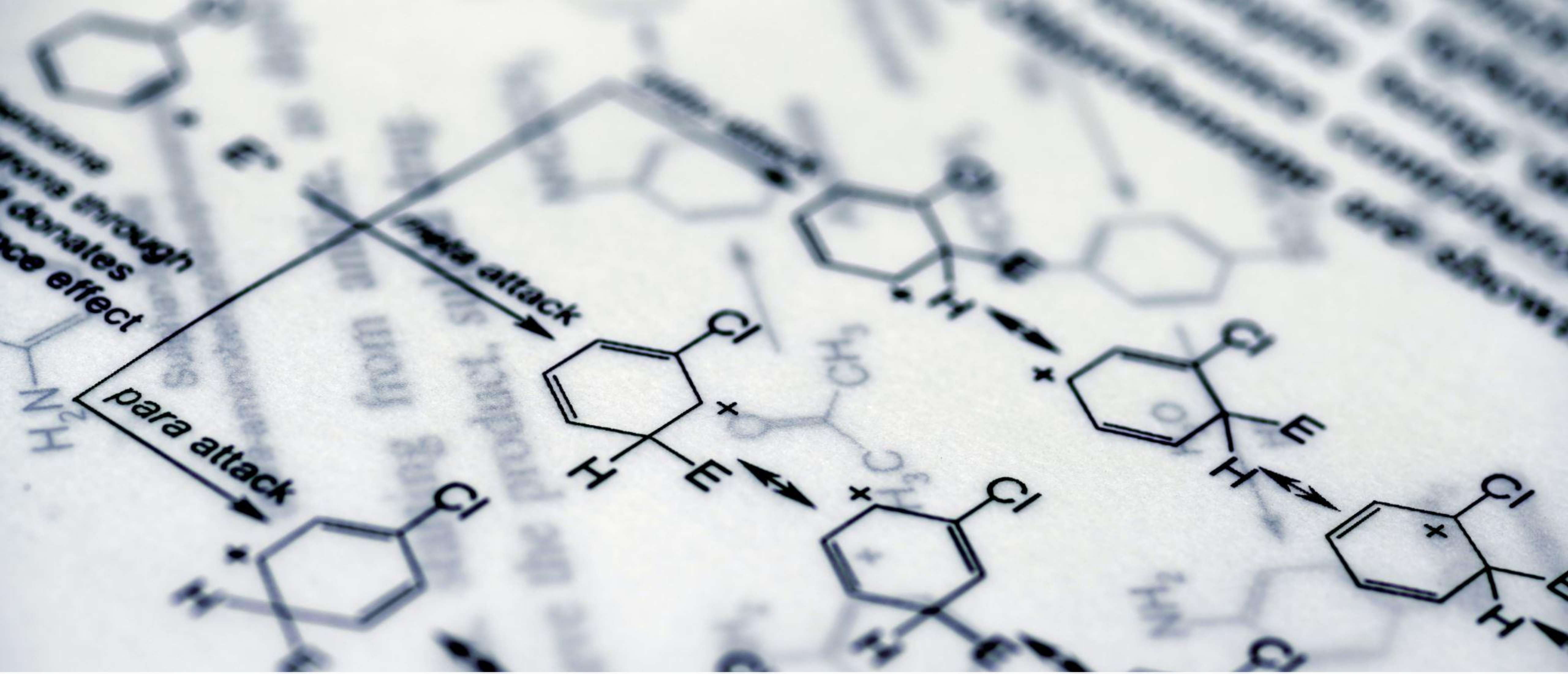
Impacto de los DOACS

	NOAC (events)	Warfarin (events)	RR (95% CI)	p
RE-LY[5]*	134/6076	199/6022	0·66 (0·53–0·82)	0·0001
ROCKET AF[6]†	269/7081	306/7090	0·88 (0·75–1·03)	0·12
ARISTOTLE[7]‡	212/9120	265/9081	0·80 (0·67–0·95)	0·012
ENGAGE AF–TIMI 48[8]§	296/7035	337/7036	0·88 (0·75–1·02)	0·10
Combined (random)	911/29 312	1107/29 229	0·81 (0·73–0·91)	<0·0001



DOAC vs Warfarina: reducción hasta del **27%** de stroke.

Ruff C, Giugliano R, Braunwald E et al. Comparison of the efficacy and safety of new oral anticoagulants with warfarin in patients with atrial fibrillation: a meta-analysis of randomised trials. The Lancet, 2013; 383, 955-962



Características de los anticoagulantes

Metabolismo DOACS

	Dabigatran[106,376]	Apixaban[517]	Edoxaban[518]	Rivaroxaban[519,520]
Bioavailability	3–7%	50%	62%	15 mg/20 mg: 66% without food, 100% with food
Prodrug	Yes	No	No	No
Clearance non-renal/renal of absorbed dose	20%/80%	73%/27%	50%/50%	65%/35%
Plasma protein binding	35%	87%	55%	95%
Dialysability	50–60% (In part dialysable)	14% (Not dialysable)	NA (Not dialysable)	NA (Not dialysable)
Metabolism	Glucoronic acid conjugation	CYP3A4 (25%), CYP1A2, CYP2J2, CYP2C8, CYP2C9 CYP2C19	CYP3A4 (<4% of elimination)	CYP2A4 (18%)[519], CYP2J2
Absorption with food	No effect	No effect	6–22% more; minimal effect on exposure	+39% more (see above)
Absorption with H2B/PPI	−12% to 30% (not clinically relevant)	No effect	No effect	No effect
Time to peak levels (h)	3	3	2–4	2–4
Elimination half-life (h)	12–17	12	10–14	5–9 (young) 11–13 h (elderly)

62%

• ≈100% (with food)

Steffel J, Collins R, Antz M, Cornu P, Desteghe L, Haeusler KG, Oldgren J, Reinecke H, Roldan-Schilling V, Rowell N, Sinnaeve P, Vanassche T, Potpara T, Camm AJ, Heidbüchel H; External reviewers. 2021 European Heart Rhythm Association Practical Guide on the Use of Non-Vitamin K Antagonist Oral Anticoagulants in Patients with Atrial Fibrillation. Europace. 2021 Oct 9;23(10):1612-1676.

WARFARIN: PHARMACOLOGY & CLINICAL SUMMARY

1. MECHANISM OF ACTION

Warfarin competitively inhibits VKORC1 (vitamin K epoxide reductase complex subunit 1)
→ decreases regeneration of active vitamin K
→ reduces γ-carboxylation and hepatic synthesis of vitamin K–dependent proteins.

Warfarin ⊣ VKORC1 → ↓ Active vitamin K

↓ Synthesis of:
- Factors II, VII, IX, X
- Proteins C & S

Protein C falls first → transient prothrombotic state at initiation (consider bridging in high-risk patients).

2. PHARMACODYNAMICS

- Warfarin is a racemic mixture of (R)- and (S)-enantiomers.
- (S)-enantiomer is 3–5 times more potent than (R)-enantiomer.

Factor / Protein	Approx. Half-life	Impact on Warfarin Effect
Factor VII	~ 6 h	INR begins to rise 24–36 h
Protein C	~ 8 h	
Factor IX	~ 24 h	Clinical anticoagulation ~5 days
Factor X	~ 36 h	
Factor II (Prothrombin)	~ 50–72 h	Maximal effect 5–7 days

Onset of action: 24–72 h | Peak effect: 5–7 days
Duration of action: 2–5 days after discontinuation

3. PHARMACOKINETICS

Absorption	• Rapid and complete • Tmax ~4 h
Distribution	• Vd ~ 0.14 L/kg • 99% protein bound (albumin)
Metabolism	• Hepatic metabolism, mainly CYP2C9 • Minor: CYP2C8, 2C18, 2C19, 1A2, 3A4 • Genetic variants (CYP2C9): – *1/*2 or *1/*3 → ~37% ↓ clearance (S-warfarin) – *2/*2, *2/*3 or *3/*3 → ~70% ↓ clearance (S-warfarin) • t½: 20–60 h (highly variable)
Elimination	• Primarily eliminated as metabolites via glomerular filtration (92%) • Not removed by hemodialysis

4. DOSING & INR TARGETS

Administration

- **Oral tablets**: 1, 2, 2.5, 3, 4, 5, 6, 7.5, 10 mg
- Once daily in the afternoon or evening
- Dose highly individual – based on metabolism, diet (vitamin K), genetics, comorbidities, drug interactions, adherence, etc.

Indication	Target INR (Range)
Venous thromboembolism	2.5 (2.0–3.0)
Non-valvular atrial fibrillation	2.5 (2.0–3.0)
Mechanical heart valves	
• Bileaflet aortic (sinus rhythm, no LA enlargement)	2.5 (2.0–3.0)
• Bileaflet or tilting disk (mitral position)	3.0 (2.5–3.5)
• Caged ball or caged disk	3.0 (2.5–3.5)
Bioprosthetic mitral valve (first 3 months)	2.5 (2.0–3.0) (consider longer if AF, prior TE, or LV dysfunction)
Post–myocardial infarction (high risk)	INR 2.0–3.0 + Aspirin ≤100 mg/day for at least 3 months

Special Populations

- **Hepatic impairment**: may be more sensitive → close INR monitoring
- **Renal impairment:** no dose adjustment; monitor closely
- **Pregnancy**: Category D (mechanical valves) / Category X (others); crosses placenta
- **Breastfeeding**: safe (not excreted in milk); monitor infant
- **Older adults:** ↑ bleeding risk → monitor closely, consider lower doses
- **Diet:** keep vitamin K intake consistent; avoid cranberry juice & limit alcohol; grapefruit juice & alcohol ↑ anticoagulant effect

5. ADVERSE EFFECTS

Most serious: Bleeding & major hemorrhage (eg, intracranial, GI bleeding, hematemesis, intraocular, hemarthrosis)

Other: easy bruising, nausea, vomiting, abdominal pain, bloating, flatulence, altered taste

Rare but Important

- Purple toe syndrome (3–8 weeks after initiation)
- Warfarin-induced skin necrosis (early; due to protein C deficiency; higher risk with PC/PS deficiency)
- Calciphylaxis (with or without ESRD)

6. DRUG INTERACTIONS (Examples)

Can ↑ bleeding risk or ↓ anticoagulant effect.

- Antiplatelet agents
- Fibrinolytics
- NSAIDs
- Antimicrobials
- Antiarrhythmic drugs (eg, amiodarone)
- Other anticoagulants

⚠ (S)-warfarin interactions are most significant (3–5x more potent). Consult interaction database.

7. CONTRAINDICATIONS (Major)

- Hypersensitivity to warfarin or components
- Significant hemorrhage risk
- Active GI ulceration or bleeding; bleeding from respiratory or GU tracts
- CNS bleeding; dissecting aortic aneurysm
- Recent or upcoming surgery of eye, CNS, or major traumatic surgery
- Epidural/spinal puncture or regional/lumbar block anesthesia
- Pericarditis with effusion or bacterial endocarditis with bleeding
- Threatened abortion, preeclampsia, or eclampsia
- Unsupervised patients with high risk of nonadherence
- Malignant hypertension
- Pregnancy (except mechanical heart valves at high thromboembolic risk)

8. MONITORING

- Monitor INR (preferred over PT): healthy baseline ≈ 1.0
- Most patients: target INR 2.0–3.0 (see section 4 for exceptions)
- Monitoring frequency: daily when starting (inpatient); then every 4 weeks when stable; more often if out of range, or with interacting drugs/dose changes
- Monitor for bleeding: bruising, hematomas, epistaxis, melena, hematuria, etc.
- Baseline Hgb/Hct; recheck every 6 months
- Consider LFTs, renal function, stool occult blood as indicated

9. TOXICITY & REVERSAL

Suspect with bleeding or supratherapeutic INR (especially >5.0).

Management

1. Hold warfarin
2. Vitamin K (phytonadione):
   - PO preferred if no major bleeding
   - IV if needed (INR should ↓ within 24 h)
3. Severe bleeding or very high INR:
   - 4-factor PCC preferred
   - or FFP ± Vitamin K
   - Activated Factor VII (if needed)

Evaluación de riesgo de thrombosis e indicación de anticoagulación en FA

Modelos de predicción de riesgo de TE

	Sub-Topic	ACC/AHA/ACCP/HRS	CCS/CHRS	CAEP	ESC	SEMES/SEC/SETH
Prevention of stroke and systemic embolism	Stroke risk assessment tools	A validated clinical risk score Eg, CHA_2DS_2-VASc, ATRIA, GARFIELD	CHADS-65	CHADS-65	In the absence of other locally validated alternatives: CHA_2DS_2-VA	CHA_2DS_2-VA
	Tool criteria for anticoagulation	Eg, CHA_2DS_2-VASc Males ♂: – 0 pt: does not qualify – 1 pt: consider – 2 pt: qualifies Females ♀: – 0-1 pt: does not qualify – 2 pt: consider – 3 pt: qualifies	If any factor in the algorithm is present (save solely vascular disease): qualifies		– 0 pt: does not qualify – 1 pt: consider – 2 pt: qualifies	
	How committee determined eligibility for an anticoagulant	▪ Low risk: <1%/y ▪ Intermediate: 1 - <2%/y ▪ High: ≥2%/y – High risk: Yes – Intermediate risk: “Reasonable” to give, consider other RFs (eg, CRF, obesity, etc) in decision – Low risk: No	<1.5%/y risk: No ≥1.5%/y risk: Yes		*Similar to ACC/AHA/ACCP/HRS	

Atzema CL, Cox JL, Cheung CC, Coll-Vinent B, Benjamin EJ, Jackevicius CA, Vinson DR. Comparing Guidelines for Atrial Fibrillation: Focus on Emergency Medicine. Ann Emerg Med. 2026 Apr;87(4):435-450.

Limitaciones principales

- Se desarrolló y validó en **cohortes ambulatorias con FA estable**.
- No incluye pacientes con **causas corregibles o secundarias.**
- No contempla **estado inflamatorio ni activación protrombótica aguda.**
- Peso **arbitrario** de variables.

Evaluación de riesgo

- Sustratos del paciente.
- Riesgo tromboembólico **objetivo.**
- Contraindicaciones para la anticoagulación.

Evaluación individual de riesgo trombótico

CHA₂DS₂-VA Score Calculator

2024 ESC Guidelines for Atrial Fibrillation

- Chronic Heart Failure (1 point)[1,2,3]
- Hypertension (1 point)[4,5]
- Age 65-74 years (1 point)
- Age ≥ 75 years (2 points)[6]
- Diabetes Mellitus (1 point)[7]
- Stroke / TIA (2 points)
- Vascular Disease (1 point)[8,9,10]

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Stroke Risk With and Without Anticoagulation

Annual stroke risk for patients with atrial fibrillation, based on their CHA₂DS₂-VA score:

Score	Annual Risk without OAC	Estimated stroke risk while on NOAC*	Risk Category
0	**0.5%**	**~0.2%**	**Very Low**
1	1.5%	~0.5%	Low
2	2.9%	~1.0%	Moderate
3	5.1%	~1.8%	Moderate
4	7.3%	~2.6%	High
5	11.2%	~3.9%	High
6	15.5%	~5.4%	Very High
7	14.7%	~5.1%	Very High
8	19.5%	~6.8%	Very High

Evaluación del riesgo de sangrado HAS-BLED

Hypertension Uncontrolled, >160 mmHg systolic	No 0	Yes +1
Renal disease Dialysis, transplant, Cr >2.26 mg/dL or >200 µmol/L	No 0	Yes +1
Liver disease Cirrhosis or bilirubin >2x normal with AST/ALT/AP >3x normal	No 0	Yes +1
Stroke history	No 0	Yes +1
Prior major bleeding or predisposition to bleeding	No 0	Yes +1
Labile INR Unstable/high INRs, time in therapeutic range <60%	No 0	Yes +1
Age >65	No 0	Yes +1
Medication usage predisposing to bleeding Aspirin, clopidogrel, NSAIDs	No 0	Yes +1
Alcohol use ≥8 drinks/week	No 0	Yes +1

Espectro de Interacciones Farmacológicas (DDI)

Riesgo Tromboembólico

Disminución de niveles plasmáticos del NACO.

Contraindicado o extrema precaución

Ejemplos:
Inductores P-gp/CYP3A4 (ej. Rifampicina, Fenitoína, Carbamazepina).

Zona de Seguridad

Niveles terapéuticos óptimos.

Monitoreo estándar

Ejemplos:
Sin interacciones relevantes.

Riesgo de Sangrado

Aumento de niveles plasmáticos del NACO.

Reducir dosis o Contraindicado

Ejemplos:
Inhibidores P-gp/CYP3A4 (ej. Ketoconazol, inhibidores proteasa VIH).

Indicaciones y contraindicaciones

Condition	Eligibility for NOAC	Comment
Mechanical prosthetic valve	Contraindicated	Excluded from pivotal RCTs Data indicating worse outcome[15,16]
Moderate to severe mitral stenosis (usually rheumatic)	Contraindicated	Excluded from pivotal RCTs Little rationale for less efficacy and safety vs. VKA
Other mild to moderate valvular disease (e.g. degenerative aortic stenosis, mitral regurgitation etc.)	Included in NOAC trials	Data regarding efficacy and safety overall consistent with patients without valvular heart disease[12,17–22]
Bioprosthetic valve/valve repair (after >3 months postoperative)	Acceptable	Some data from NOAC RCTs Single RCT indicating non-inferiority to VKA[24] Patients without AF usually on ASA after 3–6 months post-surgery, hence NOAC therapy acceptable for stroke prevention if diagnosed with AF
Severe aortic stenosis	Limited data (excluded in RE-LY)	No pathophysiological rationale for less efficacy and safety Most will undergo intervention
Transcatheter aortic valve implantation	Acceptable	Single RCT + observational data May require combination with APT[25,26]
Percutaneous transluminal aortic valvuloplasty	With caution	No prospective data May require combination with APT
Hypertrophic cardiomyopathy	Acceptable	No rational for less efficacy and safety vs. VKA Observational data positive for NOACs[33–36]

Steffel J, Collins R, Antz M, Cornu P, Desteghe L, Haeusler KG, Oldgren J, Reinecke H, Roldan-Schilling V, Rowell N, Sinnaeve P, Vanassche T, Potpara T, Camm AJ, Heidbüchel H; External reviewers. 2021 European Heart Rhythm Association Practical Guide on the Use of Non-Vitamin K Antagonist Oral Anticoagulants in Patients with Atrial Fibrillation. Europace. 2021 Oct 9;23(10):1612-1676.

Recomendaciones y aspectos por considerar

Recommendations	Class[a]	Level[b]
Direct oral anticoagulants are recommended in preference to VKAs to prevent ischaemic stroke and thromboembolism, except in patients with mechanical heart valves or moderate-to-severe mitral stenosis.[25–28,292–294]	I	A
A target INR of 2.0–3.0 is recommended for patients with AF prescribed a VKA for stroke prevention to ensure safety and effectiveness.[295–298]	I	B
Switching to a DOAC is recommended for eligible patients that have failed to maintain an adequate time in therapeutic range on a VKA (TTR <70%) to prevent thromboembolism and intracranial haemorrhage.[299–303]	I	B
Keeping the time in therapeutic range above 70% should be considered in patients taking a VKA to ensure safety and effectiveness, with INR checks at appropriate frequency and patient-directed education and counselling.[304–308]	IIa	A
Maintaining VKA treatment rather than switching to a DOAC may be considered in patients aged ≥75 years on clinically stable therapeutic VKA with polypharmacy to prevent excess bleeding risk.[309]	IIb	B
A reduced dose of DOAC therapy is not recommended, unless patients meet DOAC-specific criteria,[c] to prevent underdosing and avoidable thromboembolic events.[310–312]	III	B

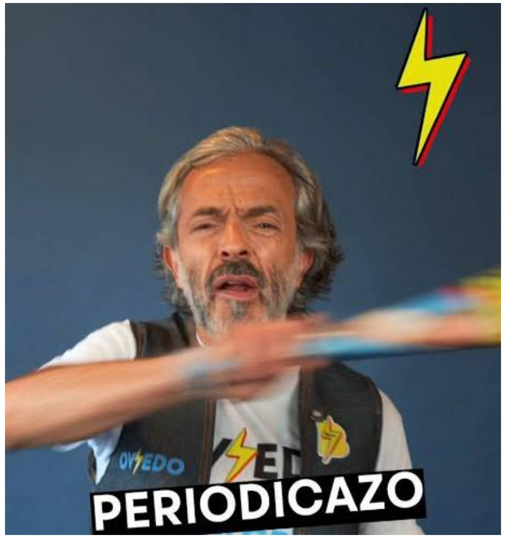
Van Gelder IC. et al. ESC Scientific Document Group. 2024 ESC Guidelines for the management of atrial fibrillation developed in collaboration with the European Association for Cardio-Thoracic Surgery (EACTS). Eur Heart J. 2024 Sep 29;45(36):3314-3414. Erratum in: Eur Heart J. 2025 Nov 3;46(41):4349.

Dosis y ajustes recomendados

Stroke prevention in atrial fibrillation (SPAF)		
	Standard dose	**Comments/dose reduction**
Apixaban[47]	5 mg BID	2.5 mg BID if two out of three fulfilled: weight ≤60 kg, age ≥80 years, serum creatinine ≥133 µmol/L (1.5 mg/dL) (*or* single criterion: if CrCl 15–29 mL/min)
Dabigatran[48]	150 mg BID/110 mg BID	No pre-specified dose-reduction criteria in phase III trial[a]
Edoxaban[49]	60 mg QD	30 mg QD if: weight ≤60 kg *or* CrCl 15–49 mL/min *or* concomitant therapy with strong P-Gp inhibitor (see 'Pharmacokinetics and drug-drug interactions of NOACs' section)
Rivaroxaban[46]	20 mg QD	15 mg QD if CrCl ≤15–49 mL/min

Steffel J, Collins R, Antz M, Cornu P, Desteghe L, Haeusler KG, Oldgren J, Reinecke H, Roldan-Schilling V, Rowell N, Sinnaeve P, Vanassche T, Potpara T, Camm AJ, Heidbüchel H; External reviewers. 2021 European Heart Rhythm Association Practical Guide on the Use of Non-Vitamin K Antagonist Oral Anticoagulants in Patients with Atrial Fibrillation. Europace. 2021 Oct 9;23(10):1612-1676.

¡OJO!



Treatment of DVT/PE

	Initial therapy	Remainder of treatment phase
Apixaban[498]	10 mg BID, 7 days	5 mg BID, no dose reduction
Dabigatran[499]	Heparin/LMWH	150 mg BID, no dose reduction[a]
Edoxaban[500]	Heparin/LMWH	60 mg QD, same dose reduction as for SPAF (see above)
Rivaroxaban[501,502]	15 mg BID, 21 days	20 mg QD, no dose reduction[b]

Steffel J, Collins R, Antz M, Cornu P, Desteghe L, Haeusler KG, Oldgren J, Reinecke H, Roldan-Schilling V, Rowell N, Sinnaeve P, Vanassche T, Potpara T, Camm AJ, Heidbüchel H; External reviewers. 2021 European Heart Rhythm Association Practical Guide on the Use of Non-Vitamin K Antagonist Oral Anticoagulants in Patients with Atrial Fibrillation. Europace. 2021 Oct 9;23(10):1612-1676.

Metanálisis DOACS.

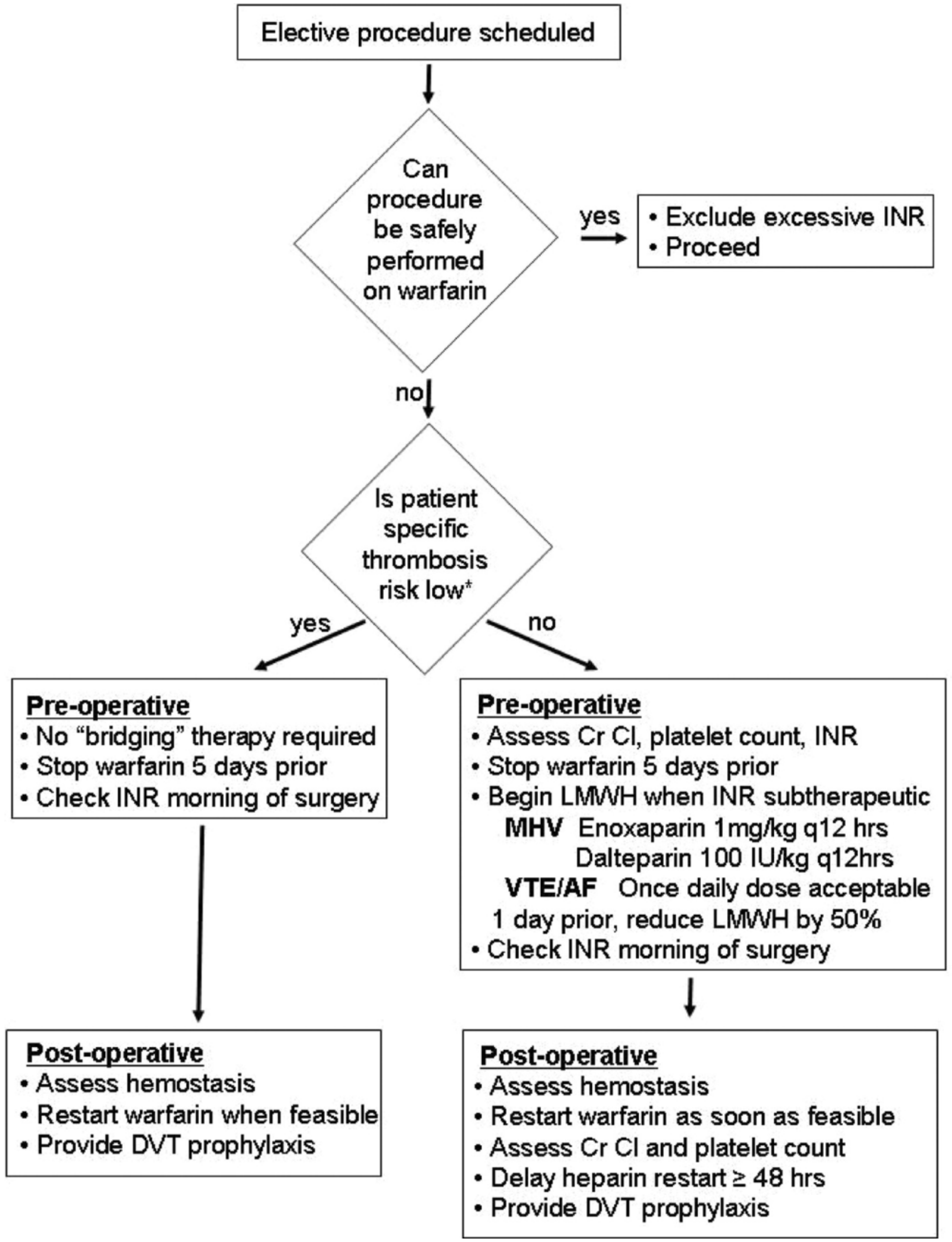
Outcome	Comparison	Number of studies (N)	Pooled Relative Risk (95% CI)
Total mortality	apixaban vs. dabigatran	11 (515,706)	1.03 (0.93, 1.15)
	apixaban vs. rivaroxaban	13 (1,267,040)	**0.86 (0.79, 0.95)**
Stroke/systemic embolism	apixaban vs. dabigatran	12 (828,396)	**0.89 (0.81, 0.98)**
	apixaban vs. rivaroxaban	14 (1,826,473)	0.92 (0.85, 1.00)
Ischemic stroke	apixaban vs. dabigatran	18 (394,554)	**0.85 (0.73, 0.99)**
	apixaban vs. rivaroxaban	19 (1,087,791)	**0.75 (0.58, 0.98)**
Intracranial hemorrhage	apixaban vs. dabigatran	15 (922,271)	0.96 (0.85, 1.10)
	apixaban vs. rivaroxaban	18 (1,973,314)	**0.76 (0.68, 0.85)**
Major bleeds	apixaban vs. dabigatran	16 (375,591)	**0.80 (0.72, 0.89)**
	apixaban vs. rivaroxaban	15 (592,394)	**0.62 (0.53, 0.74)**

Perry T, editor. Therapeutics Letter. Vancouver (BC): Therapeutic Initiatives. Letter 146, Apixaban is safer and more effective than rivaroxaban for non-valvular atrial fibrillation. 2023 Dec.

Switch hacia otros anticoagulantes

Steffel J, Collins R, Antz M, Cornu P, Desteghe L, Haeusler KG, Oldgren J, Reinecke H, Roldan-Schilling V, Rowell N, Sinnaeve P, Vanassche T, Potpara T, Camm AJ, Heidbüchel H; External reviewers. 2021 European Heart Rhythm Association Practical Guide on the Use of Non-Vitamin K Antagonist Oral Anticoagulants in Patients with Atrial Fibrillation. Europace. 2021 Oct 9;23(10):1612-1676.

Terapia puente



Wysokinski WE, McBane RD 2nd. Periprocedural bridging management of anticoagulation. Circulation. 2012 Jul 24;126(4):486-90.

Recurrencia de trombosis

Recommendation	Class[a]	Level[b]
A thorough diagnostic work-up should be considered in patients taking an oral anticoagulant and presenting with ischaemic stroke or thromboembolism to prevent recurrent events, including assessment of non-cardioembolic causes, vascular risk factors, dosage, and adherence.[356,357]	IIa	B
Adding antiplatelet treatment to anticoagulation is not recommended in patients with AF to prevent recurrent embolic stroke.[356,359]	III	B
Switching from one DOAC to another, or from a DOAC to a VKA, without a clear indication is not recommended in patients with AF to prevent recurrent embolic stroke.[252,356,359]	III	B

Van Gelder IC. et al. ESC Scientific Document Group. 2024 ESC Guidelines for the management of atrial fibrillation developed in collaboration with the European Association for Cardio-Thoracic Surgery (EACTS). Eur Heart J. 2024 Sep 29;45(36):3314-3414. Erratum in: Eur Heart J. 2025 Nov 3;46(41):4349.

Gestión de Crisis: Matriz de Sangrado y Reversión

Definición	Acción
Sangrado menor, epistaxis, hematomas.	Retrasar la siguiente dosis o suspender temporalmente. Identificar el origen. Re-evaluar interacciones farmacológicas. Intervención local.

Definición	Acción y Agentes de Reversión
Hemorragia intracraneal, sangrado gastrointestinal masivo, inestabilidad hemodinámica.	Soporte hemodinámico inmediato. Suspender NACO.
	• **Para Dabigatrán:** Administrar Idarucizumab (5g IV).
	• **Para Inhibidores del Factor Xa** (Apixabán, Rivaroxabán): Administrar Andexanet alfa.
	• **Alternativa:** Usar Concentrados de Complejo Protrombínico (PCC). No usar plasma fresco congelado (PFC).

INR sobreterapéutico

SIN SANGRADO:

INR < 5: Diferir dosis de anticoagulante.

INR 5 – 10: omitir 1 u 2 dosis, control con INR.

INR < 10: podría considerar vitamina K.

CON SANGRADO:

Cualquier INR: suspender anticoagulante, vitamina K, PCC o PFC.

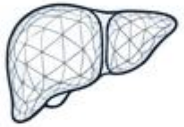
Escenarios especiales

Depuración Metabólica: Ajustes por Disfunción Orgánica

Riñón

Calcular aclaramiento de creatinina (CrCl) exclusivamente con fórmula Cockcroft-Gault

- **CrCl > 50 ml/min:** Dosis estándar (vigilar edad/peso).
- **CrCl 15-50 ml/min:** Considerar reducción de dosis según ficha técnica.
- **CrCl < 15 ml/min o Diálisis:** NACOs no recomendados/contraindicados.



Hígado

Evaluar función hepática mediante clasificación Child-Pugh

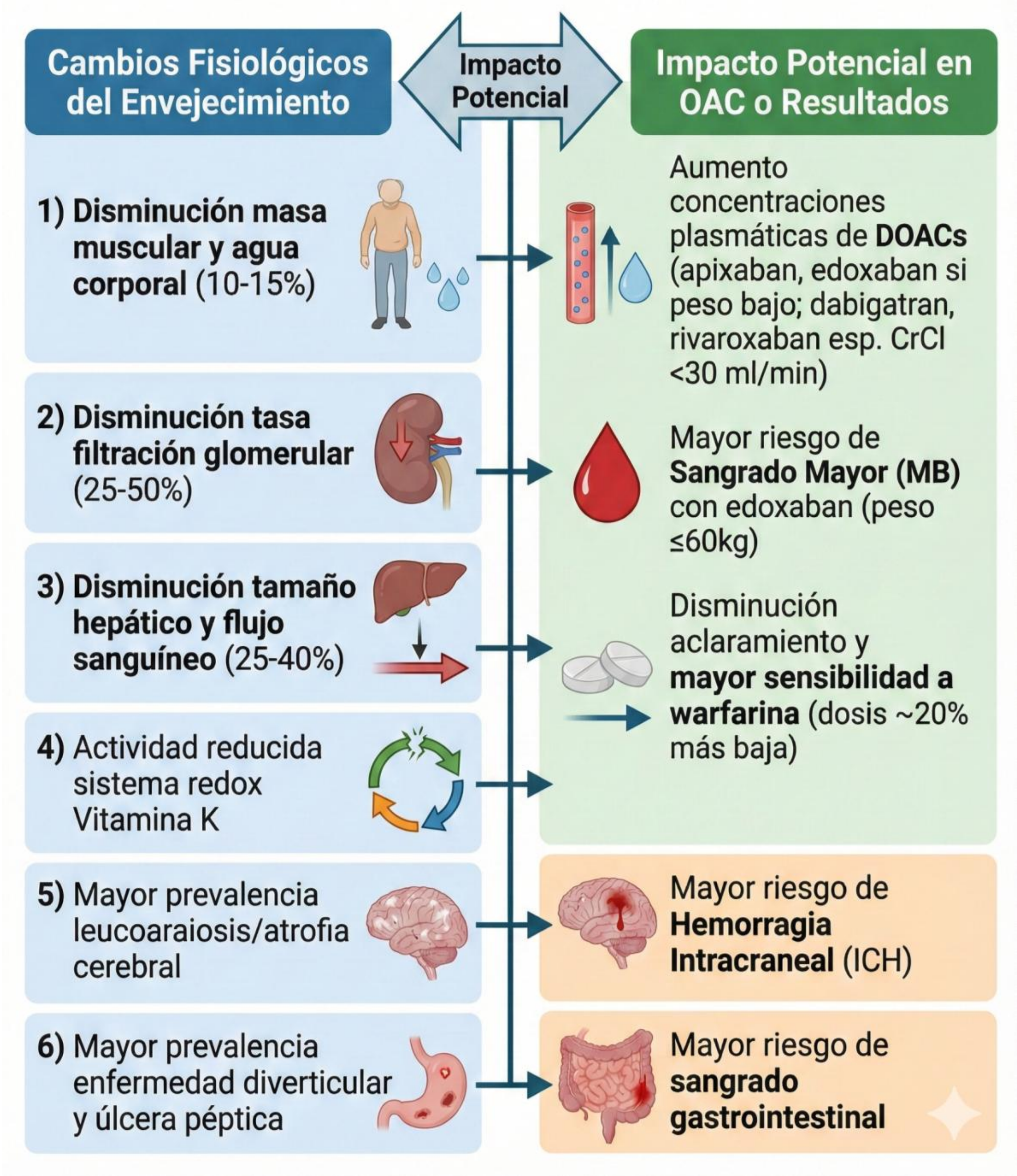
- **Child-Pugh A:** Uso seguro para todos los NACOs.
- **Child-Pugh B:** Usar con precaución. (Rivaroxabán no recomendado).
- **Child-Pugh C (o con coagulopatía):** Todos los NACOs contraindicados.

Obesidad

	Obesity (BMI > 30kg/m^2)	Low body weight (BMI < 18.5kg/m^2)
AF	• Obesity may be protective against stroke • BMI < 40kg/m^2 – any DOAC • BMI > 40kg/m^2 – apixaban or rivaroxaban favoured • BMI > 50kg/m^2 – apixaban or rivaroxaban strongly favoured, consider dose monitoring	• Strongly increases risk of stroke and mortality • Likely to signify other aspects of frailty • Apixaban and edoxaban most supported by evidence • Dose adjustments required for apixaban*, edoxaban

Karakasis P, Ktenopoulos N, Pamporis K, Sagris M, Soulaidopoulos S, Gerogianni M, Leontsinis I, Giannakoulas G, Tousoulis D, Fragakis N, Tsioufis K. Efficacy and Safety of Direct Oral Anticoagulants versus Warfarin in Obese Patients (BMI ≥ 30 kg/m^2) with Atrial Fibrillation or Venous Thromboembolism: An Updated Systematic Review and Meta-Analysis. J Clin Med. 2024 Jun 27;13(13):3784.

Cambios fisiológicos del envejecimiento



J. Stuby, M. Haschke, T. Tritschler, D. Aujesky, Oral anticoagulant therapy in older adults, Thrombosis Research, Volume 238. 2024, Pages 1-10.

Caídas

- Un modelo analítico de decisión de Markov (1999) demostró que un paciente tendría que caerse **295 veces** para que el riesgo de un hematoma subdural superara el beneficio de la anticoagulación con VKA.
- Dado el menor riesgo de hemorragia intracraneal con NOAC en comparación con VKA, el **“número necesario de caídas” sería incluso mayor con NOAC.**

LAS CAÍDAS NO SON UNA CONTRAINDICACION DE ANTICOAGULACIÓN SE DEBEN MANEJAR LOS FACTORES DE RIESGO PARA DISMINUIR LAS CAIDAS

Europace. 2021;23(10):1612–1676. Doi:10.1093/europace/euab06

ARISTOPHANES (Pacientes frágiles con FANV)

- Estudio observacional retrospectivo.
- Medicare + 3 bases de datos comerciales.
- Se crearon 6 cohortes emparejadas por propensity score.
- Seguimiento: 6–8 meses promedio.
- 150.487 pacientes frágiles (34% de toda la cohorte de >65 años).Alta comorbilidad: 90% con CHA_2DS_2-VASc ≥480% con HAS-BLED ≥3Edad promedio: 83–84 años.

- **Apixabán:** menor riesgo de ACV/ES y sangrado mayor vs. warfarina.
- **Dabigatrán:** menor riesgo de sangrado mayor; eficacia similar.
- **Rivaroxabán:** menor ACV/ES pero mayor riesgo de sangrado mayor.
- Todos los NOAC reducen hemorragia intracraneal.
- **Apixabán tiene el mejor perfil global (eficacia + seguridad).**

Lip GYH, Keshishian A, Li X, Hamilton M, Masseria C, Gupta K, Luo X, Mardekian J, Friend K, Nadkarni A, Pan X, Baser O, Deitelzweig S. Effectiveness and Safety of Oral Anticoagulants Among Nonvalvular Atrial Fibrillation Patients. Stroke. 2018 Dec;49(12):2933-2944. Erratum in: Stroke. 2020 Feb;51(2):e44.

Estrategia Antitrombótica Post-ICP: Escalonamiento Terapéutico

1-7 Días (Alta hospitalaria)

Terapia Triple

(NACO + Aspirina + Clopidogrel)

*Preferir Clopidogrel sobre Ticagrelor.

Hasta 6-12 meses

Terapia Dual

(NACO + Clopidogrel)

*Suspender aspirina temprano.

Post 1 año

Monoterapia

(Solo NACO)

Des-intensificar: Riesgo alto de sangrado, riesgo aterotrombótico bajo.

Prolongar combinación: Stent en tronco común, bifurcación proximal, trombosis previa, riesgo bajo de sangrado.

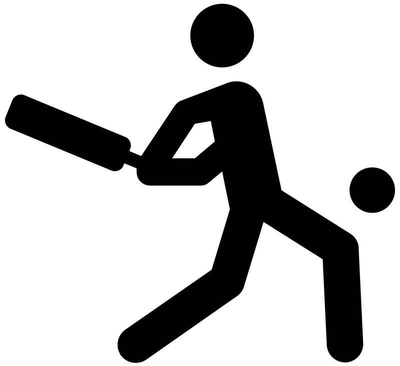
Evitar stents metálicos (BMS). Usar IBP (Protectores gástricos) siempre en terapia triple/dual.

Fibrilación auricular aguda (de novo)

AHA SCIENTIFIC STATEMENT

Atrial Fibrillation Occurring During Acute Hospitalization: A Scientific Statement From the American Heart Association

In the absence of contraindications to oral anticoagulation, long-term oral anticoagulation according to stroke risk "may be reasonable" on the basis of the high risk of recurrence for patients with newly-diagnosed atrial fibrillation in the context of acute triggers.



The NEW ENGLAND JOURNAL of MEDICINE

ESTABLISHED IN 1812 SEPTEMBER 15, 2011 VOL. 365 NO. 11

Apixaban versus Warfarin in Patients with Atrial Fibrillation

The NEW ENGLAND JOURNAL of MEDICINE

ESTABLISHED IN 1812 SEPTEMBER 8, 2011 VOL. 365 NO. 10

Rivaroxaban versus Warfarin in Nonvalvular Atrial Fibrillation

- Los pacientes con fibrilación auricular con desencadenantes "reversibles" han sido excluidos rutinariamente de los ensayos clínicos de anticoagulantes orales.
- Ausencia de evidencia directa de ECA en este contexto.

Conclusiones

- La indicación de anticoagulación depende del riesgo tromboembólico, no de la frecuencia ni del tipo de FA.
- Warfarina sigue siendo indispensable en escenarios específicos, especialmente prótesis valvulares mecánicas y ciertas valvulopatías.
- Suspender anticoagulación por miedo al sangrado también tiene costo clínico: el riesgo tromboembólico debe valorarse de forma equilibrada.
- La mejor anticoagulación no es la más potente: es la que el paciente puede recibir de forma segura y sostenida.

Gracias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